

2025 AMC 10A Problem 5 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 5. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 5

Problem:

Consider the sequence of positive integers

$$1, 2, 1, 2, 3, 2, 1, 2, 3, 4, 3, 2, 1, 2, 3, 4, 5, 4, 3, 2, 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1, 2, \dots$$

What is the 2025th term in this sequence?

Answer Choices:

- A. 5
- B. 15
- C. 16
- D. 44
- E. 45

Solution:

We observe the sequence forms blocks as follows:

$1 \rightarrow 1, 2,$
 $2 \rightarrow 1, 2, 3, 2,$
 $3 \rightarrow 1, 2, 3, 4, 3, 2,$
 $4 \rightarrow 1, 2, 3, 4, 5, 4, 3, 2,$ and so on.

Each n^{th} block contains $2n$ terms.

The total number of terms up to the n^{th} block is

$$2(1 + 2 + 3 + \cdots + n) = 2 \times \frac{n(n+1)}{2} = n(n+1).$$

We require that

$$n(n+1) \leq 2025.$$

This gives

$$n^2 + n - 2025 \leq 0.$$

Approximating n , we get

$$n \approx 44 \quad \text{since} \quad 44 \times 45 = 1980 < 2025 < 45 \times 46 = 2070.$$

Hence, the 44^{th} sequence has 88 terms and the 45^{th} sequence has 90 terms.

The 2025^{th} term lies in the 45^{th} sequence, and it is the

$$2025 - 1980 = 45^{\text{th}} \text{ term of that sequence.}$$

In the 45th sequence

$$1, 2, 3, \dots, 46, 45, \dots, 2,$$

the 45th term is clearly

(E) 45

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

Powered by [Wiki.js](#)